

Unemployment Analysis with Python

Project Overview

This project was completed as part of the CodeAlpha Data Science Internship. The objective of this project is to analyze unemployment trends in India using Python. It focuses on data cleaning, exploratory data analysis (EDA), visualization, and understanding the impact of Covid-19 on unemployment.

Two unemployment datasets were used to study employment patterns across different states and regions of India.

Objectives

The main objectives of this project are:

- Analyze unemployment rate data.
- Perform data cleaning and preprocessing.
- Explore unemployment trends through visualizations.
- Study the impact of Covid-19 on unemployment.
- Compare state-wise and region-wise unemployment patterns.
- Generate meaningful insights from the data.

Datasets Used

1. Unemployment in India
2. Unemployment Rate upto 11_2020

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Data Cleaning

The datasets were prepared by:

- Checking missing values
- Removing null values

- Verifying column names and datatypes
- Understanding dataset structure

Data Visualization

The following visualizations were used:

- Histogram
- Bar Plot
- Box Plot
- Heatmap
- Line Plot

These visualizations helped in identifying unemployment trends and understanding the impact of Covid-19 more effectively.

Covid-19 Impact Analysis

The project examines unemployment trends during the Covid-19 period. The analysis indicates that unemployment rates increased considerably in many states and regions during the pandemic.

Trend analysis and line plots were used to visualize these changes over time.

Key Insights

- Some states recorded higher unemployment rates than others.
- Covid-19 significantly affected employment across India.
- Unemployment patterns varied across regions.
- Monthly unemployment trends fluctuated during the pandemic.
- Data visualization helped uncover important employment trends.

Conclusion

This project analyzed unemployment trends in India using Python and various data analysis libraries. The project involved data cleaning, visualization, trend analysis, and Covid-19 impact analysis.